

Why network monitoring?

Since we're today well aware of the importance of network monitoring within any organisation, let's have a look at some of the reasons for this:

1. *Helps us plan for changes:* Network monitoring solutions allow us to study any constant problem in more detail. For instance, if some hardware keeps constantly tripping, we may need to replace it. The same applies to a service that crashes repeatedly.
2. *Keeps you informed:* With a real-time monitoring system, if any failure or irregularity is detected, it can be immediately communicated by different means such as SMS, emails, pagers or a network message. Hence, we will be notified of any problem on our network, wherever we may be, which allows us to fix the issue swiftly. Without any network monitoring solution, we would have to look for issues on our own, or wait for the issue to be reported to us, to work towards a solution.
3. *Reports issues:* Network monitoring reports can help us spot specific trends in the system's performance. These highlight the need for any upgrades or replacements, and document the otherwise 'unseen' work which keeps the IT systems we manage, running smoothly.
4. *Diagnoses issues:* Imagine a scenario where one of your company's websites goes down. Without network monitoring, you may not be able to even tell if the problem is related to just the website, the Web server or the applications on which the website runs. Network monitoring will actually pinpoint the specific point of failure, saving your time and money, which you would otherwise have had to spend to diagnose the problem.
5. *Remedies disasters:* If you are immediately notified that there is some issue with one of your systems on a network, and the issue might take quite some time to fix, then the time saved by being alerted immediately can be actually used to bring in a backup system that can replace the current failure, thereby providing an efficient service to your customers. Some network monitoring solutions can even automatically move to correct the problem caused by restarting a service (or multiple services) upon failure.
6. *Keeps track of your Web application:* Many services that companies offer to their users or customers are actually just Web applications running on a server. Network monitoring solutions allow you to stay on top of different website problems, spot issues before users or customers notice them, and remedy those issues in a timely fashion.
7. *Ensures the efficient operation of security systems:* Although businesses spend a lot of money, resources and time on security hardware or software, without a network monitoring solution, they cannot be really sure that the security devices are working as expected. Network monitoring solutions can effectively monitor and manage the health of such critical software and hardware security

systems. With the help of another feature that this product offers, i.e., patch management, we can also streamline the automation and management of different Microsoft software updates and patches.

8. *Fixes problems, anytime, anywhere:* Nowadays, network monitoring products are being shipped with different remote access features. They offer just one-click remote support for any server or workstation in your environment. Apart from providing a much faster service, remote access also helps in saving a lot of money without having to roam to branch offices or customer sites.
9. *Saves money:* Network monitoring products help in fixing issues faster with instant alerts, spot small and big issues, and eliminate the need for any manual checks on different event logs, backup systems, hard disks, antivirus systems and other devices. All this facilitates cost saving as well as revenue building.
10. *Ensures uptime:* Network monitoring maximises network availability by monitoring all systems on your network, including workstations, servers and network devices or applications. Whenever a failure is detected, you will immediately be notified via the alerts that you configure in the product, allowing you to take corrective action in a highly efficient manner.

Monitoring security aspects of the network

It's so important for IT administrators to be able to react as quickly as possible in order to protect a system from potential malware attacks. If installed antivirus systems and firewalls don't discover these attacks in time, then the damage done can even bring all operations to a standstill. At that time, administrators will just be able to react to these problems, instead of being able to proactively take measures to prevent these problems before they occur. The fact is that these firewalls and virus scanners alone are not always sufficient to ensure the all-around security of the network. Companies that integrate a network monitoring solution in their security strategy are able to discover these potential dangers to the company network at early stages in the following ways.

1. Network monitoring solutions help to check the existing security systems, such as firewalls and virus scanners, for reliability. For example, the monitoring solution gathers detailed data regarding the performance as well as status of the firewall, around the clock. If the firewall is not working properly, then the risk of a malware attack on the network becomes high. To avoid this, the administrators are informed of abnormalities in the firewall at the early stages.
2. The monitoring software also checks different virus scanners running on the central mail server. This helps different companies to make sure that the scanner is continuously active. The monitoring solution even uses special sensors to check the Windows Security Center in